# LED Power Consumption

How the EHS Production Tool calculates LED wall power & current

This note explains, end to end, how the tool turns a panel's rated wattage into the power and amperage figures you see on the LED tab and in the project PDF. Every value below can be overridden per screen.

## Panel wattages (current catalog)

These are the rated (peak, full-white) wattages per cabinet in the tool today. The 0.5 x 1 m cabinet exists in two rows — bare and with its captive cable loom — but both pull the same power.

| Cabinet | Size (m) | Pixels | Rated power |
|---|---|---|---|
| Uniview UR Pro 0.5x1m | 0.5 x 1.0 | 128 x 256 | 350 W |
| Uniview UR Pro 0.5x1m + cable | 0.5 x 1.0 | 128 x 256 | 350 W |
| Uniview UR Pro 0.5x0.5m 90 deg | 0.5 x 0.5 | 128 x 128 | 175 W |
| Uniview UR Pro 0.5x0.5m 90 deg + cable | 0.5 x 0.5 | 128 x 128 | 175 W |
| Custom panel (fallback) | 0.5 x 0.5 | 128 x 128 | 175 W |

## The build-up, step by step

1. **Start with the panel's rated power**

   Every cabinet in the catalog has a rated wattage (its peak draw running full white). This is the starting point for one cabinet.

2. **Scale it for brightness**

   You rarely run full brightness. Nominal calibration point is 5000 nits. Even at 0 nits a panel still pulls 30% of rated power (receiving cards, fans, processors never switch off) — that is the floor. Above the floor it scales linearly up to 100% at nominal.  Formula: brightnessFactor = 0.30 + 0.70 x (yourNits / 5000), capped at 1.0. Example: 2500 nits = 0.30 + 0.70 x 0.5 = 65%.

3. **Add PSU overhead**

   Power supplies lose energy and you want a safety margin, so a percentage is added on top (default 25%, overridable per screen).  Formula: wattsPerCabinet = panel.power x brightnessFactor x (1 + overhead/100).

4. **Multiply by enabled cabinets**

   Only cabinets that are actually powered count — disabled or blanked ones are excluded.  Formula: totalWatts = wattsPerCabinet x (number of enabled cabinets).

5. **Convert watts to amps**

   Using the region's mains voltage (e.g. 230 V EU, 120 V US) and the power factor (default 0.95).  Formula: amps = totalWatts / (voltage x powerFactor).

6. **Split across power chains**

   If you set a max-cabinets-per-chain, the tool works out how many chains you need and divides the amps evenly. A chain is flagged as overloaded when it exceeds 80% of the breaker rating (the standard de-rate).

## Worked example — a 5 m x 3 m wall

A wall 5 m wide x 3 m tall (physical size). With the Uniview UR Pro 0.5 x 1 m cabinet (0.5 m wide x 1.0 m tall) that is 10 cabinets wide x 3 tall = 30 cabinets, all powered. At full brightness, 25% PSU overhead, 230 V EU mains, 0.95 power factor.

| Step | Calculation | Result |
|---|---|---|
| Wall -> cabinets | (5.0 / 0.5) x (3.0 / 1.0) | 10 x 3 = 30 |
| Brightness factor | full (5000 nits) -> 1.0 | 1.0 |
| Watts per cabinet | 350 x 1.0 x 1.25 | 437.5 W |
| Total (Max output) | 437.5 x 30 | 13,125 W |
| Current | 13125 / (230 x 0.95) | 60.1 A |
| Average output | 13125 / 3 | 4,375 W |
| Chains (6/chain) | ceil(30 / 6) | 5 chains |
| Amps per chain | 60.1 / 5 | 12.0 A |

Read-out: this 5 x 3 m wall needs about 60 A total at full white — across 5 power chains that is ~12 A each, just inside a 16 A breaker's 80% ceiling (12.8 A). For generator and heat planning, expect roughly 4.4 kW average on real content.

Dim the wall and it drops fast: the same 5 x 3 m wall at 2500 nits uses brightness factor 0.65, so watts per cabinet = 350 x 0.65 x 1.25 = 284 W, total = 8,531 W (Max), and current = 39.0 A.

## Max output vs Average output

**Max output = totalWatts above — peak draw, full white. Size your cable, distro and breakers off THIS number.**

Average output = Max x 1/3. Real content (video, text, logos) almost never lights every pixel white, so realistic running draw is roughly a third of peak. Use it for generator fuel and heat planning — never for sizing power infrastructure.

## Signal & power chaining (rigging limits)

Beyond raw load, two practical limits decide how you cable a wall: how many cabinets you can daisy-chain on one data link, and how many on one power feed. The figures below are for the Uniview UR Pro 0.5 x 1 m cabinet at 50 Hz refresh, fed by a 16 A powerCON TRUE1 via a Soca breakout mounted at the top of the screen.

**Signal — panels per data link**

| Colour depth | Panels per data link (50 Hz) |
|---|---|
| 10-bit | up to 18 |
| 8-bit | up to 24 |

Lower colour depth carries more panels per port because each pixel costs fewer bits of the port's fixed bandwidth budget. These are 50 Hz figures (standard in Norway); at 60 Hz the counts drop by roughly a sixth.

**Power — panels per TRUE1 chain (top-fed)**

| Daisy-chain on one 16 A TRUE1 | Panels | Approx. current |
|---|---|---|
| Comfortable design link | 9 | ~14.4 A |
| Hard maximum | 10 | ~16.0 A |

At nameplate 350 W per cabinet, each draws ~1.6 A (230 V, 0.95 PF). Nine leaves headroom; ten sits right at the 16 A connector limit at full white. Feeding from the top means the top cabinet's connector carries the whole chain's current, so it sets the ceiling.

Note: the LED tab is deliberately more cautious — about 6 cabinets per chain — because it adds 25% PSU overhead and an 80% breaker de-rate on top of the nameplate. That is the safe continuous design target; 9-10 is the physical hardware ceiling.

## What you can override per screen

Voltage region - Power factor (default 0.95) - PSU overhead % (default 25%) - Brightness in nits - Cabinets per power chain. Each has a sensible default but can be set per screen.

## In one line

**Rated cabinet watts -> dim it for brightness -> pad it for PSU losses -> multiply by cabinet count -> divide by volts to get amps -> split across chains and check against breakers.**